

Poster: Enabling High-Fidelity and Real-Time Mobility Digital Twin with Edge Computing

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Abstract—A Mobility Digital Twin is an emerging implementation of Digital Twin in the transportation domain, and has been attracting extensive attention from both industry and academia. Although a few research have been conducted on the mobility digital twin, there is no systematic work with an end-to-end digital twin model construction framework. In this paper, we propose an end-to-end system framework, including sensory data collection, offloading, and processing, that aims to facilitate a high-fidelity and real-time digital twin model construction for connected and automated vehicles. Additionally, preliminary experiments are conducted to demonstrate our research motivation and to guide the future system framework design.

Index Terms—edge computing, extended reality, digital twin

I. BACKGROUND AND MOTIVATION

The recent development of edge computing, Artificial Intelligence (AI), and wireless communications have been reshaping the automotive industry. For example, in 2019, Toyota announced a profound transformation from being an automaker to become a mobility company, with an emphasis on Connectivity, Autonomous driving, Shared mobility, and Electrification of vehicles (CASE) [1]. The implementation of the CASE proves to be a key element to realize the vision of a safe, sustainable, frictionless, and personalized mobility society. One of the killer applications of the CASE would be Mobility Digital Twin [2].

The Mobility Digital Twin (MDT) is an emerging implementation of Digital Twin (DT) in the transportation domain, where it aims to create a perpetual digital replica of a human driver or a connected and automated vehicle (CAV) in the cyber world based on the data acquired in the physical world. The MDT is attracting increasing attention from both industry and academia. According to a market research report, the global DT market size was valued at USD 7.48 billion in 2021 and is projected to grow at a compound annual growth rate of 39.1% from 2022 to 2030, where the automotive and transportation industry accounted for the largest revenue share of more than 19.0% of the overall revenue [3]. In addition, a few MDT research have been conducted and demonstrate the MDT has great potentials to facilitate a variety of connected mobility services, such as personalized adaptive cruise control (P-ACC) [4], cooperative automated driving [5], and autonomous driving simulation with a 3D DT environment [6].

One of the intrinsic features shared with these connected mobility services is that they require continuous and rich sensor data collected in the physical world to construct or maintain a *high-fidelity* and *real-time* DT model for each participant (e.g., human driver and car). For instance, a huge amount of data can be generated by various types of perception sensors (e.g., LiDAR, radar, and stereo camera) deployed on CAVs (e.g., 4 terabytes of raw data in two hours [7]). Therefore, *designing an end-to-end framework to efficiently perform the data collection, offloading, and processing is crucial to the high-fidelity and real-time DT model construction*. However, to the best of our knowledge, there is no existing end-to-end framework developed for the DT model construction.

II. INITIAL DESIGN

Our research proposes an end-to-end adaptive DT model construction framework with edge computing, which aims to assure the high-fidelity and real-time digital replicas of physical entities, e.g., 3D DT model construction of vehicles. The proposed framework consists of three major components: CAV, DT model generation pipeline, and edge server, as illustrated in Fig. 1.

CAV. CAVs are the contributors as well as the customers of the DT model construction. In practical scenarios, it is inadequate to build a high-fidelity DT model by collecting sensory data from the limited number of roadside infrastructures only. Therefore, in our proposed framework, the data collection can be performed in a crowdsourcing manner, where each CAV can actively contribute to the data collection and cooperatively construct the DT model with other entities such as other CAVs or an edge server.

DT Model Generation Pipeline. The pipeline is designed to construct a 3D DT model by leveraging the physical world sensory data input from CAVs. In this work, the input data are stereo images collected by CAVs moving on the road, and will be processed through three key intermediate phases: disparity map generation, point cloud generation, and 3D mesh generation.

Edge Server. The edge server provides the capabilities of real-time data processing and pre-built DT model storage. In this work, the edge server is built upon KubeEdge, an open-source system for extending native containerized application

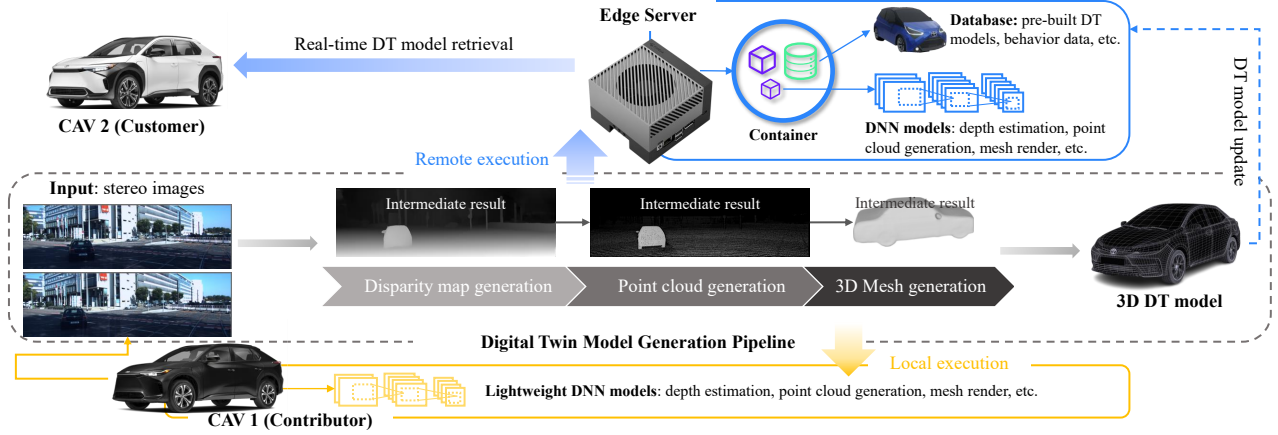


Fig. 1. Overview of the proposed end-to-end adaptive DT model construction framework with edge computing.

TABLE I
COMPARISON OF DIFFERENT PHASES RUNNING ON EDGE SERVER (TESLA V100) IN TERMS OF INFERENCE LATENCY, ACCURACY, OFFLOADED DATA SIZE, ETC.

DT Generation Pipeline Phase	Inference Latency (s)	Accuracy*	Offloaded Data Size (MB)	Input & Output	Model
Disparity Map Generation	0.160	98.35%	1.58	Stereo images & Disparity map	ACVNet [8]
	0.040	97.85%	1.58	Stereo images & Disparity map	BGNet [9]
Point Cloud Generation	0.239	81.6%	0.38	Disparity map & Point Cloud	Lyu et al. [10]
3D Mesh Generation	0.233	78%	0.025	Point Cloud & Mesh	Lionar et al. [11]

*Experimental data sets: KITTI [12] and ShapeNet [13]

orchestration capabilities to hosts at edge. In addition, as highlighted in Fig. 1, our proposed DT model construction framework allows the DT model generation pipeline to be *fully or partially processed at either the CAV or the edge server*. Therefore, an adaptive processing scheme is indispensable to dynamically determine the DT model construction strategy based on the DT model fidelity requirement and real-time network conditions.

III. PRELIMINARY RESULTS

In this section, we conduct preliminary experiments to study the inference latency, data volume, and accuracy of each phase in the DT model generation pipeline. The experimental study can help us to understand the computation and communication resource requirements of each phase, and guide the design of the adaptive processing scheme.

Table I presents the detailed information of the tested models, input & output data, and results (e.g., average inference latency, accuracy, and the amount of data required to be offloaded). All the experiments are conducted on the edge server, an Nvidia Tesla V100 GPU. For disparity map generation, models such as BGNet [9] that are specially tuned for real-time purpose can achieve approximately 40 ms inference latency. Although ACVNet [8] offers a slightly better accuracy, it is not feasible in the real-time DT model construction due to its large latency. Furthermore, considering the large offloaded data size, it may not be worth executing the disparity map generation remotely when the network condition is poor.

In addition, we observe that the point cloud and 3D mesh generation are more computation-intensive, and have a less offloaded data size. Note that, for different processing phases, the accuracy is measured with different criteria. For example, the accuracy of disparity map generation is calculated by the percentage of accurate pixels in total [14]; the accuracy of point cloud generation is evaluated by mean intersection-over-union (mIoU) [15]; and the accuracy of 3D mesh generation is quantified by F-score [16].

Those preliminary results advocate designing an end-to-end adaptive DT model construction framework as:

- 1) The total processing latency (i.e., from stereo images to 3D mesh) is far from achieving a real-time DT model construction. Although our experiments are running on a data center GPU with state-of-the-art algorithms, the average end-to-end latency is still over 0.5 s.
- 2) Interactions between DT model configurations (e.g., fidelity) and performance criterion are complex and lack analytic understandings.

IV. SUMMARY AND NEXT STEPS

In this paper, we proposed an end-to-end adaptive DT model construction framework with edge computing to facilitate the high-fidelity and real-time MDT. Research motivation and preliminary results were presented to support our initial design of the framework. The next step is to tackle the proposed two challenges in Section III and complete the system and adaptive processing scheme design.

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